

10 Mathematical Induction

Prove the following statements with either induction, strong induction or proof by smallest counterexample.

2. For every integer $n \in \mathbb{N}$, it follows that $1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 1$, this statement is $1^2 = \frac{(1)(2)(3)}{6} = \frac{6}{6}$, which is true.
2. Assume $1^2 + 2^2 + 3^2 + 4^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$ for some integer $n = k \geq 1$.
Observe that this implies the statement is true for $n = k + 1$.

$$\begin{aligned}
 &1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \\
 &(1^2 + 2^2 + 3^2 + \dots + k^2) + (k+1)^2 = \\
 &\quad \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + \frac{6(k+1)^2}{6} \\
 &= \frac{k(k+1)(2k+1) + 6(k+1)(k+1)}{6} \\
 &= \frac{2k^3 + 9k^2 + 13k + 6}{6} \\
 &= \frac{(k+1)(k+2)(2k+3)}{6} \\
 &= \frac{(k+1)([k+1]+1)(2[k+1]+1)}{6}
 \end{aligned}$$

Therefore, $1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{(k+1)([k+1]+1)(2[k+1]+1)}{6}$, which means the statement is true for $n = k + 1$. $\square$

4. If $n \in \mathbb{N}$, then $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + 4 \cdot 5 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 1$, this statement is $1(1+1) = \frac{1(1+1)(1+2)}{3} = 2$, which is true.
2. Assume $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$ for some integer $n = k \geq 1$. Observe that this implies the statement is true for $n = k+1$.

$$\begin{aligned}
 1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1) + (k+1)(k+2) &= \\
 (1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1)) + (k+1)(k+2) &= \\
 \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) &= \frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3} \\
 &= \frac{k(k^2 + 3k + 2) + 3(k^2 + 3k + 2)}{3} \\
 &= \frac{k^3 + 6k^2 + 11k + 6}{3} \\
 &= \frac{(k+1)(k^2 + 5k + 6)}{3} \\
 &= \frac{(k+1)(k+2)(k+3)}{3} \\
 &= \frac{(k+1)([k+1]+1)([k+1]+2)}{3}
 \end{aligned}$$

Therefore $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + k(k+1) + (k+1)(k+2) = \frac{(k+1)([k+1]+1)([k+1]+2)}{3}$, which means the statement is true for $n = k+1$. $\square$